

状态 = 3, 5, 8

3.

状态空间 {0, 1, 2, ...}

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$$P_{ij} = \begin{cases} p & j = i+1 \\ 1-p & j = 0 \end{cases}$$

5.

41

$$Y_0 = 1 \Rightarrow X_1 = X_2 = 1$$

$$Y_1 = 6 \Rightarrow X_3 = 6 \Rightarrow Y_2 = 6$$

$$P(Y_2 = 1 | Y_0 = 1, Y_1 = 6) = 0$$

$$P(Y_1 = 6) = \frac{1+5+5}{36} = \frac{11}{36}$$

$$P(Y_2 = 1 | Y_1 = 6) = \frac{P(Y_2 = 1, Y_1 = 6)}{P(Y_1 = 6)}$$

$$= \frac{36}{11} \times P(X_2 = 6, X_3 = 1, X_4 = 1)$$

$$= \frac{36}{11} \times \frac{1}{6} \times \frac{1}{6} \times \frac{1}{6}$$

$$= \frac{1}{66}$$

$$(2) \quad Z_0 = 2 \Rightarrow X_1 = X_2 = 1, \quad Z_1 = 7 \Rightarrow X_3 = 6$$

$$Z_2 = 12 \Rightarrow X_4 = 6$$

$$P(Z_2 = 12 | Z_0 = 2, Z_1 = 7) = \frac{1}{6}$$

$$P(Z_2 = 12 | Z_1 = 7) = \frac{P(X_2 = 1, X_3 = 6, X_4 = 6)}{\frac{1}{6}} = \frac{1}{36}$$

(3)

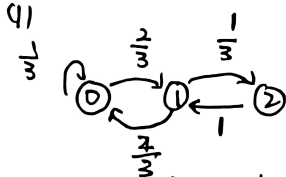
$$P(Y_2 = 1 | Y_0 = 1, Y_1 = 6) \neq P(Y_2 = 1 | Y_1 = 6)$$

$$P(Z_2 = 12 | Z_0 = 2, Z_1 = 7) \neq P(Z_2 = 12 | Z_1 = 7)$$

都不是马尔科夫链

8.

91



$$\mu_0 = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$$

$$P(X_2 = 0 | X_0 = 0) = \frac{1}{3} \times \frac{1}{3} + \frac{2}{3} \times \frac{1}{3} = \frac{5}{9}$$

$$P(X_0 = 0 | X_2 = 0) = \frac{P(X_0 = 0) P(X_2 = 0 | X_0 = 0)}{P(X_2 = 0)}$$

$$= \frac{\frac{1}{3} \times \frac{5}{9}}{\frac{1}{3} \times \frac{5}{9} + \frac{2}{3} \times \frac{1}{3}} = \frac{5}{13}$$

(2)

$$P(X_1 = 0) = \frac{1}{3} \times (\frac{1}{3} + \frac{2}{3}) = \frac{1}{3}$$

$$P(X_1 = 0, X_3 = 0, X_4 = 1, X_6 = 1)$$

$$= P(X_1 = 0) P(X_3 = 0 | X_1 = 0) P(X_4 = 1 | X_3 = 0) P(X_6 = 1 | X_4 = 1)$$

$$= \frac{1}{3} \times \frac{5}{9} \times \frac{2}{3} \times \frac{1}{3} = \frac{20}{243}$$

(3)

$$f_{11}^{(2)} = \frac{4}{9} + \frac{1}{3} = \frac{7}{9}$$

$$f_{11}^{(3)} = \frac{4}{27}$$

$$f_{11}^{(n)} = \begin{cases} \frac{7}{9} & n=2 \\ \frac{4}{27} & n \geq 3 \end{cases}$$

μ_1

$$\mu_1 = \frac{14}{9} + \sum_{n=3}^{\infty} \frac{4n}{3^n}$$

$$\frac{x}{1-x} = \sum_{n=0}^{\infty} x^{n+1} \quad |x| < 1$$

$$\left(\frac{1}{1-x}\right)' = \sum_{n=0}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$$

$$\sum_{n=3}^{\infty} \frac{n}{3^n} = \frac{3}{4} = \sum_{n=3}^{\infty} \frac{n}{3^n} + \frac{1}{9}$$

$$\mu_1 = \frac{14}{9} + \frac{2}{9} = \frac{16}{9}$$

$$f_{11} = \frac{7}{9} + 4 \times \frac{1}{3} = \frac{7}{9} + \frac{4}{3} = \frac{19}{9}$$

$$\text{综上 } f_{11}^{(n)} = \begin{cases} 0 & n=0 \\ \frac{7}{9} & n=2 \\ \frac{4}{27} & n \geq 3 \end{cases} \quad f_{11} = 1 \quad \mu_1 = \frac{7}{3}$$

1111